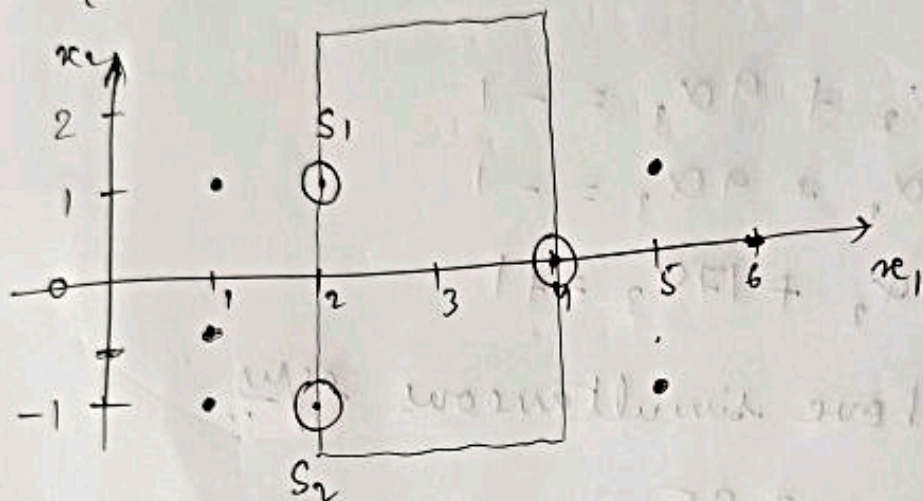


8) Find optimal hyperplane for the set of data pts. and classes.

class 1: $\{(1, 1), (2, 1), (1, -1), (2, -1)\}$
 class 2: $\{(4, 0), (5, 1), (5, -1), (6, 0)\}$



We select 3 support vectors S_1 , S_2 and S_3
 here $S_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $S_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $S_3 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$

use ^{vectors} augmented with 1 as a bias input.

$$\tilde{S}_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

$$\tilde{S}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$S_3 = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

Now to find 3 parameters α_1, α_2 and α_3 based on following 3 linear eqns

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_1 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_1 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_1 = -1 \quad (\text{-ve class})$$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_2 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_2 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_2 = -1 \quad (\text{-ve class})$$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_3 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_3 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_3 = +1 \quad (\text{+ve class})$$

Let us substitute the values for $\tilde{S}_1, \tilde{S}_2$ and $\tilde{S}_3$ in above eqns

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = +1$$

$$6\alpha_1 + 4\alpha_2 + 9\alpha_3 = -1$$

$$4\alpha_1 + 6\alpha_2 + 9\alpha_3 = -1$$

$$9\alpha_1 + 9\alpha_2 + 17\alpha_3 = +1$$

Simplifying above simultaneous eqns

$$\alpha_1 = \alpha_2 = -3.25$$

$$\alpha_3 = 3.5$$

Hyperplane that divides +ve class from -ve class is given by

$$\tilde{w} = \sum_i \alpha_i \tilde{S}_i$$

$$\tilde{w} = \alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

$$= (-3.25) \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + (-3.5) \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + (3.5) \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$$

The vectors are augmented with a bias. Hence we can equate the entry in $\tilde{w}$ as the hyperplane with an offset b .

So, separating hyperplane eqn $y = wx + b$
with $w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and offset $b = -3$.

